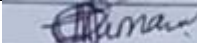
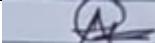
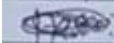
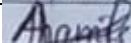


THE STATE UNIVERSITY OF ZANZIBAR (SUZA)

THE SCHOOL OF BUSINESS(SOB)



GROUP 15:

NAMES	REG NO	SIGNATURE
MUNIRA OMAR JUMA	24BIA029	
ASMA ALI ABDULLAH	24BIA068	
HASSAN MAHAMOUD AME	24BIA070	
ASIA IDDI KHAMIS	24BIA030	

Question 1.

OUTPUT:

Options
Calling Father
Calling Mother
Calling Friend
Brand:Samsung Model:Galaxy S24 Price:2500000.0 Battery:85% On:true
Battery: 100%
Brand:Samsung Model:Galaxy S24 Price:2500000.0 Battery:100% On:true

QUESTION 2.

OUTPUT:

Options
EMP001 Ali Hassan IT Salary:1500000.0 Bonus:750000.0 Net:1912500.0
EMP002 Fatma Said HR Salary:1200000.0 Bonus:240000.0 Net:1224000.0
After promotion:
EMP001 Ali Hassan Senior IT Salary:1800000.0 Bonus:900000.0 Net:2295000.0

QUESTION 3.

OUTPUT:

```
=== Product Inventory Exercise ===
```

```
[PRD001] Laptop (Electronics) | Price: 1500000.00 | Stock: 25 | Reorder: No  
[PRD002] Office Chair (Furniture) | Price: 350000.00 | Stock: 8 | Reorder: YES  
[PRD003] Notebook (Stationery) | Price: 5000.00 | Stock: 100 | Reorder: No
```

```
--- Processing Transactions ---
```

```
Selling 20 laptops: Success
```

```
Does Office Chair need reorder? true
```

```
Adding 50 units to Office Chair...
```

```
Applying 15% discount to Notebooks...
```

```
--- Final Inventory Value ---
```

```
[PRD001] Laptop (Electronics) | Price: 1500000.00 | Stock: 5 | Reorder: YES  
[PRD002] Office Chair (Furniture) | Price: 350000.00 | Stock: 58 | Reorder: No  
[PRD003] Notebook (Stationery) | Price: 4250.00 | Stock: 100 | Reorder: No
```

```
Total Warehouse Value: 28225000.00
```

QUESTION 4.

OUTPUT:

```
00PinstructorDr. AliinstructorName40creditHours3maxCapacity4enrolledStudenttrueisActive  
Available:-1  
Cannot enroll!  
Cannot enroll!  
Cannot enroll!  
Cannot enroll!  
Cannot enroll!  
Full? true  
Final info:  
00PinstructorDr. AliinstructorName40creditHours3maxCapacity1enrolledStudenttrueisActive  
DBinstructorMs. FatmainstructorName35creditHours3maxCapacity0enrolledStudenttrueisActive
```

QUESTIO 5.

OUTPUT:

```
Area      : 50.0
Perimeter : 30.0
Diagonal  : 11.180339887498949
Is Square : false
-----
Length    : 7.0
Width     : 7.0
Color     : blue
Filled    : true
Area      : 49.0
Perimeter : 28.0
Diagonal  : 9.899494936611665
Is Square : true
-----
rect1 is square: true
rect2 is square: false
rect3 is square: true

rect1 compared to rect2: smaller

rect1 after resize:
Length    : 3.0
Width     : 3.0
Color     : white
Filled    : false
Area      : 9.0
Perimeter : 12.0
Diagonal  : 4.242640687119285
Is Square : true
-----
Total area of all rectangles: 108.0

=== Exercise Complete ===
```

QUESTION 6.

OUTPUT:

```
=== Simple Calculator ===
First number: 2
Second number: 3
Operation (+,-,*,/): +
2.00 + 3.00 = 5.00
```

QUESTION 7.

OUTPUT:

```
Enter exam score (0-100): 10
Score:10 Grade:F (Fail)
Status:FAILED.
```

QUESTION 8.

OUTPUT:

```
Enter number: 10
10 x 1 = 10
10 x 2 = 20
10 x 3 = 30
10 x 4 = 40
10 x 5 = 50
10 x 6 = 60
10 x 7 = 70
10 x 8 = 80
10 x 9 = 90
10 x 10 = 100
10 x 11 = 110
10 x 12 = 120
```

QUESTION 9.

OUTPUT:

```
Guess my number (1-100):
Your guess: 50
Too low!
Your guess:
```

QUESTION 10.

OUTPUT:

```
Enter 5 numbers:
1
2
3
4
5
Numbers: [ 1 2 3 4 5 ]
Sum: 15
Average: 3.00
Max: 5, Min: 1
```

QUESTION 11.

OUTPUT:

```
Options
1.Add 2.Remove 3.View 4.Search 5.Count 6.Exit
Choice: 1
Name: JUMA

1.Add 2.Remove 3.View 4.Search 5.Count 6.Exit
Choice: 3
1. JUMA

1.Add 2.Remove 3.View 4.Search 5.Count 6.Exit
Choice:
```

QUESTION 12.

OUTPUT:

MOTORCYCLE TEST

VEHICLE DETAILS

Brand: Honda

Model: CBR500R

Year:2022

Price:TZS 15,000,000.00

Engine Size : 500 CC

Helmet Box : is Not Available

Honda CBR500R engine is ready to work now

Honda CBR500R A well performs a wheelie!

Kick starting Honda CBR500R

Honda CBR500R The engine is stop now

ELECTRIC CAR TEST

VEHICLE DETAILS

Brand: Tesla

Model: Model 3

Year:2024

Price:TZS 85,000,000.00

Doors : 4

Fuel : Electric

Tesla Model 3 electric system activated silently

A Charging Tesla Model 3 Battery: 75.0 kWh

The Battery Capacity is : 75.0 kWh

A Driving Range : 450 km

Tesla Model 3 Beep Beep!

Tesla Model 3 The engine is stop now

POLYMORPHISM CHECK

VEHICLE DETAILS

Brand: Toyota

Model: Corolla

Year:2023

Price:TZS 45,000,000.00

VEHICLE SYSTEM DEMO

CAR TEST

VEHICLE DETAILS

Brand: Toyota

Model: Corolla

Year:2023

Price:TZS 45,000,000.00

Doors : 4

Fuel : Petrol

Toyota Corolla engine is ready to work now

Toyota Corolla Beep Beep!

The trunk opened for Toyota Corolla

Toyota Corolla The engine is stop now

MOTORCYCLE TEST

VEHICLE DETAILS

Brand: Honda

Model: CBR500R

Year:2022

Price:TZS 15,000,000.00

Engine Size : 500 CC

Helmet Box : is Not Available

Honda CBR500R engine is ready to work now

Honda CBR500R A well performs a wheelie!

Kick starting Honda CBR500R

Honda CBR500R The engine is stop now

ELECTRIC CAR TEST

VEHICLE DETAILS

Brand: Tesla

Model: Model 3

Year:2024

Price:TZS 85,000,000.00

POLYMORPHISM CHECK

VEHICLE DETAILS

Brand: Toyota
Model: Corolla
Year:2023
Price:TZS 45,000,000.00
Doors : 4
Fuel : Petrol
Toyota Corolla engine is ready to work now

VEHICLE DETAILS

Brand: Honda
Model: CBR500R
Year:2022
Price:TZS 15,000,000.00
Engine Size : 500 CC
Helmet Box : is Not Available
Honda CBR500R engine is ready to work now

VEHICLE DETAILS

Brand: Tesla
Model: Model 3
Year:2024
Price:TZS 85,000,000.00
Doors : 4
Fuel : Electric
Tesla Model 3 electric system activated silently

END

SUMMARY CONCLUSION

In conclusion, the group exercise was successfully carried out with full participation from all members. It helped us understand the subject better and improved our teamwork skills. The exercise was meaningful and achieved its objectives.